SZÉCHENYI EGYETEM
UNIVERSITY OF GYŐR

Final Deliverables and Evaluation Criteria

Audi Development Camp 2026

JÁRMŰIPARI KUTATÓKÖZPONT
VEHICLE INDUSTRY RESEARCH CENTER

JÁRMŰIPARI KUTATÓKÖZPONT
VEHICLE INDUSTRY RESEARCH CENTER

SZÉCHENYI EGYETEM
UNIVERSITY OF GYŐR

Build one complete AI robotics training pipeline — end to end.

A simulated Unitree G1 (29 DoF) learns a basic movement using Reinforcement Learning — and your team explains how it could ever move to a real robot.

The goal is not a perfect robot, but is a complete, reproducible engineering story you can defend.

JÁRMŰIPARI KUTATÓKÖZPONT
VEHICLE INDUSTRY RESEARCH CENTER

SZÉCHENYI EGYETEM
UNIVERSITY OF GYŐR

Drop a ball, directly in front of the robot.

Assignment

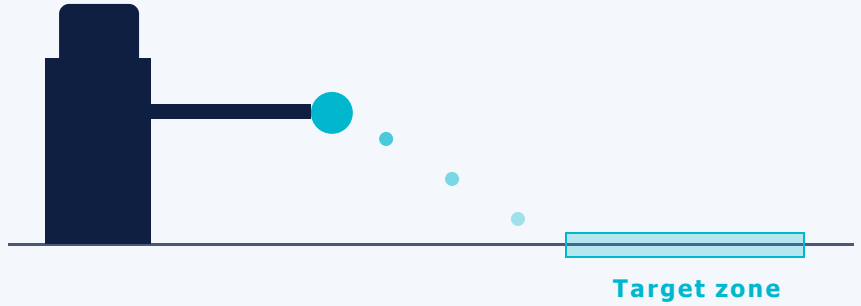
The humanoid must learn to release a ball so it lands in a **predefined target area directly in front** of the robot, while the robot maintains a **stable posture**.

The task is fixed. What differs between teams is **how** you define success, model the environment, design the reward, and read the graphs.

The target zone is not strictly defined!

The scene, in one glance

Robot stable, ball released



JÁRMŰIPARI KUTATÓKÖZPONT
VEHICLE INDUSTRY RESEARCH CENTER

SZÉCHENYI EGYETEM
UNIVERSITY OF GYŐR

Three tasks, 120 points

Each task is a phase of the same engineering story — and each is worth the same 40 points. You will be scored on how honestly you move through them, not on which one is the fanciest.

Task 1

Define & baseline

Turn the assignment into a measurable problem. Set up the sim. Build a non-learning reference behaviour.

40 points

Task 2

RL pipeline

Design the reward. Run at least one full training. Save the policy. Compare curves against your baseline.

40 points

Task 3

Sim2Real & comms

Think about what breaks on a real robot. Own your limits. Deliver the final presentation.

40 points

SZÉCHENYI EGYETEM
UNIVERSITY OF GYŐR

Task 1.

Audi Development Camp 2026

JÁRMŰIPARI KUTATÓKÖZPONT
VEHICLE INDUSTRY RESEARCH CENTER

JÁRMŰIPARI KUTATÓKÖZPONT
VEHICLE INDUSTRY RESEARCH CENTER

SZÉCHENYI EGYETEM
UNIVERSITY OF GYŐR

Define the problem

01 **A documented task definition** — ball-dropping task, assumptions, constraints.

02 **Success criteria & evaluation metrics** — how you'll know it worked.

03 **A validated simulation environment** — robot loads, steps, behaves.

04 **A non-learning baseline behaviour** — scripted, hand-tuned, whatever — but no RL.

05 **A short baseline performance summary** — how the reference actually does.

JÁRMŰIPARI KUTATÓKÖZPONT
VEHICLE INDUSTRY RESEARCH CENTER

SZÉCHENYI EGYETEM
UNIVERSITY OF GYŐR

Evaluation – Task 1.

Total
40

01
Task definition & justification
10 pts

Is the ball-drop framed as a concrete, feasible problem? Are assumptions and constraints written down?

02
Success criteria & evaluation metrics
10 pts

Can you decide, without arguing, whether one run was better than another? Are the metrics writable as numbers?

03
Simulation setup & validation
10 pts

The G1 loads, steps, and doesn't explode. Someone else can run your sim from your instructions.

04
Non-learning baseline & documentation
10 pts

A reference behaviour exists, produces numbers, and its failures are named honestly.

SZÉCHENYI EGYETEM
UNIVERSITY OF GYŐR

Task 2.

Audi Development Camp 2026

JÁRMŰIPARI KUTATÓKÖZPONT
VEHICLE INDUSTRY RESEARCH CENTER

JÁRMŰIPARI KUTATÓKÖZPONT
VEHICLE INDUSTRY RESEARCH CENTER

SZÉCHENYI EGYETEM
UNIVERSITY OF GYŐR

A runnable learning pipeline

01 **A reproducible RL pipeline** — code, structure, setup instructions.

02 **A documented reward & training config** — obs, actions, hyperparameters, rationale.

03 **At least one saved trained policy or model** — the artifact from a completed run.

04 **Evaluation graphs & baseline comparison** — curves, metrics, honest interpretation.

“Reproducible” means: your mentor can pull the repo, run one script, and reach the same graphs

JÁRMŰIPARI KUTATÓKÖZPONT
VEHICLE INDUSTRY RESEARCH CENTER

SZÉCHENYI EGYETEM
UNIVERSITY OF GYŐR

A running learning pipeline

Total
40

01
Pipeline setup & reproducibility
10 pts

Someone else can run your training from your README. Code structure makes the intent visible.

02
Reward function & training config
10 pts

The reward reflects your success criteria. Obs, actions and hyperparameters are aligned with the task.

03
Completed training run
10 pts

At least one run finishes and produces a saved policy. Logs, seeds and metadata exist alongside the model.

04
Evaluation graphs & comparison
10 pts

Curves and metrics you can defend. Trained policy is compared honestly against your baseline.

SZÉCHENYI EGYETEM
UNIVERSITY OF GYŐR

Task 3.

Audi Development Camp 2026

JÁRMŰIPARI KUTATÓKÖZPONT
VEHICLE INDUSTRY RESEARCH CENTER

JÁRMŰIPARI KUTATÓKÖZPONT
VEHICLE INDUSTRY RESEARCH CENTER

SZÉCHENYI EGYETEM
UNIVERSITY OF GYŐR

Results, Sim2Real Concept

01 **A Sim2Real gap analysis** — concrete assumptions that would fail in the real world.

02 **A robustness concept** — how you'd make the policy less fragile.

03 **A written limitation analysis** — what your solution can't do, and why.

04 **The final presentation** — task → baseline → RL → results → Sim2Real → next steps.

JÁRMŰIPARI KUTATÓKÖZPONT
VEHICLE INDUSTRY RESEARCH CENTER

SZÉCHENYI EGYETEM
UNIVERSITY OF GYŐR

Results, Sim2Real Concept

Total
40

01
Sim2Real gap analysis
10 pts

Named, specific assumptions that would fail on the real Unitree G1 — not generic disclaimers.

02
Robustness concept
10 pts

A concrete idea for making the policy less brittle — domain randomization, noise, safety envelope, etc.

03
Limitation analysis
10 pts

You name what your solution can’t do, when it fails, and what would need to change.

04
Final presentation & communication
10 pts

A clear engineering story with graphs, a defended argument, and next-step recommendations.

JÁRMŰIPARI KUTATÓKÖZPONT
VEHICLE INDUSTRY RESEARCH CENTER

SZÉCHENYI EGYETEM
UNIVERSITY OF GYŐR

Suggested deliverables

01

Working simulation project

The MuJoCo project a mentor can run.

02

Ball-drop task & baseline reference

Documented, with a non-learning behaviour.

03

A trained RL policy

Saved model or checkpoint, loadable.

04

Training evaluation graphs

Curves, metrics, axes readable.

05

Baseline vs. learned comparison

Same metric on both sides. Honest.

06

Sim2Real strategy

Specific gaps + robustness plan.

07

Final presentation

Story: task → sim → RL → risks.

08

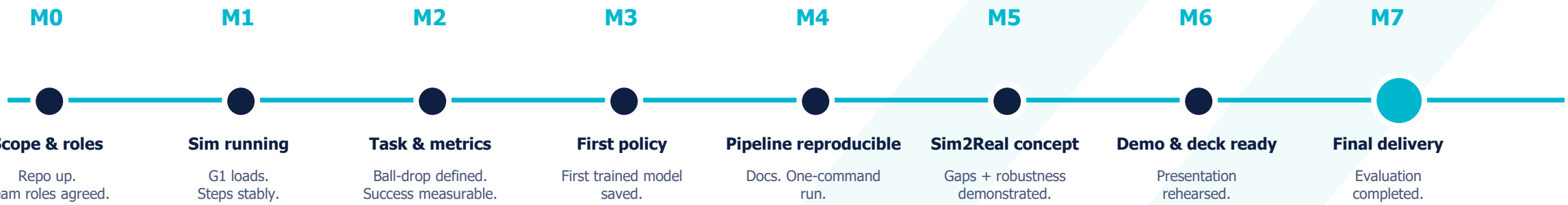
Structured project files

Predictable layout. Findable everything.

JÁRMŰIPARI KUTATÓKÖZPONT
VEHICLE INDUSTRY RESEARCH CENTER

SZÉCHENYI EGYETEM
UNIVERSITY OF GYŐR

Recommended milestones



JÁRMŰIPARI KUTATÓKÖZPONT
VEHICLE INDUSTRY RESEARCH CENTER

SZÉCHENYI EGYETEM
UNIVERSITY OF GYŐR

Constraints

FREE HAND

You choose — and defend

- RL algorithm (PPO, SAC, anything you can justify)
- Reward shaping strategy
- Observation and action space design
- What "successful drop" means in numbers
- Target area size, ball properties, release strategy
- Libraries, tooling, training compute
- How you visualise your results

HARD WALLS

Non-negotiable

- Robot: Unitree G1 (29 DoF version)
- Task: drop a ball in front of the robot (ball – 4 cm, 50 g)
- A non-learning baseline must exist
- At least one RL training run must complete
- A saved policy must be handed over
- A Sim2Real analysis is required
- Simulation only — no physical robot execution

SZÉCHENYI EGYETEM
UNIVERSITY OF GYŐR

Good luck!

Workspace setup help:
https://robotlabor.github.io/humanoid-robot-fundamentals/workspace_setup/

JÁRMŰIPARI KUTATÓKÖZPONT
VEHICLE INDUSTRY RESEARCH CENTER